

Lab 1 Solution

```
module arrays_lab;

// -----
// 1. Array Declarations
// -----
bit [7:0] arr1 [5];

// arr2: Packed array of 3 elements, each 8 bits
bit [2:0][7:0] arr2;

// arr3: Unpacked array of vectors, size 3, each element is 8 bits
bit [7:0] arr3 [3];

// arr4: Dynamic array of 8-bit elements (no initial allocation)
bit [7:0] arr4 [];

initial begin

    $display("-----");
    $display(" SystemVerilog Arrays and Looping Lab ");
    $display("-----");

    // -----
    // Task 2: Iterate using 'for' over arr1 and initialize with 2*i
    // -----
    $display("\n--- Task 2: Unpacked Array (arr1) ---");
    $display("Initializing arr1 (size: %0d) with double its index...", $size(arr1));

    for (int i = 0; i < 5; i++) begin
        arr1[i] = i * 2;
        $display(" arr1[%0d] = %0d", i, arr1[i]);
    end

    // -----
    // Task 3: Assign 24-bit hex max value to arr2 and display each 8-bit slice
    // -----
    $display("\n--- Task 3: Packed Array (arr2) ---");
    arr2 = 24'hFF_FF_FF;

    $display("arr2 (size: %0d) assigned value: %0h", $size(arr2), arr2);
    for (int i = 0; i < $size(arr2); i++) begin
        $display(" arr2[%0d] = %0h", i, arr2[i]);
    end

    // -----
    // Task 4: Initialize arr3 with hex values AA, BB, CC and display
    // -----
    $display("\n--- Task 4: Unpacked Array of Vectors (arr3) ---");
    arr3 = '{8'hAA, 8'hBB, 8'hCC};

    $display("arr3 (size: %0d):", $size(arr3));
    for (int i = 0; i < $size(arr3); i++) begin
        $display(" arr3[%0d] = %0h", i, arr3[i]);
    end

    // -----
    // Task 5: Allocate arr4 with 5 elements, copy from arr1, display with foreach
    // -----
    $display("\n--- Task 5: Dynamic Array (arr4) - Allocate & Copy ---");
    arr4 = new[5];
    $display("arr4 allocated with size: %0d", $size(arr4));

    arr4 = arr1;
    $display("Copied arr1 into arr4.");

    $display("Contents of arr4 (using foreach loop):");
    foreach (arr4[idx]) begin
        $display(" arr4[%0d] = %0d", idx, arr4[idx]);
    end
end
```

```

// -----
// Task 6: Resize arr4 by adding 2 new elements, assign values, display
// -----
$display("\n--- Task 6: Dynamic Array (arr4) - Resize ---");
arr4 = new[$size(arr4) + 2](arr4);
$display("arr4 resized to: %0d", $size(arr4));

arr4[5] = 99;
arr4[6] = 100;

$display("Contents of arr4 after resize (using for loop):");
for (int i = 0; i < $size(arr4); i++) begin
    $display("  arr4[%0d] = %0d", i, arr4[i]);
end

// -----
// Task 7: Delete the last element and display new size
// -----
$display("\n--- Task 7: Dynamic Array (arr4) - Delete Last Element ---");
if ($size(arr4) > 0) begin
    arr4 = new[$size(arr4) - 1](arr4);
    $display("arr4 size after deleting last element: %0d", $size(arr4));
end else begin
    $display("arr4 is already empty. Cannot delete more elements.");
end

// -----
// Task 8: Delete all elements using .delete() and display size
// -----
$display("\n--- Task 8: Dynamic Array (arr4) - Delete All ---");
arr4.delete();
$display("arr4 size after .delete(): %0d", $size(arr4));

#30ns;
$stop;
end

endmodule

```

Do file:

```

# =====
# ===== Cleaning and creating work library =====
# =====
vdel -all
vlib work

# =====
# ===== Compilation =====
# =====
vlog arrays_lab.sv

# =====
# ===== Simulation =====
# =====
vsim -voptargs+=acc work.arrays_lab -l Lab1_Sim.log

# =====
# ===== Running =====
# =====
add wave * ;#Or use "do wave.do" to load waveform configs
run -all

```

Log:

```
# -----  
# SystemVerilog Arrays and Looping Lab  
# -----  
#  
# --- Task 2: Unpacked Array (arr1) ---  
# Initializing arr1 (size: 5) with double its index...  
#   arr1[0] = 0  
#   arr1[1] = 2  
#   arr1[2] = 4  
#   arr1[3] = 6  
#   arr1[4] = 8  
#  
# --- Task 3: Packed Array (arr2) ---  
# arr2 (size: 3) assigned value: ffffff  
#   arr2[0] = ff  
#   arr2[1] = ff  
#   arr2[2] = ff  
#  
# --- Task 4: Unpacked Array of Vectors (arr3) ---  
# arr3 (size: 3):  
#   arr3[0] = aa  
#   arr3[1] = bb  
#   arr3[2] = cc  
#  
# --- Task 5: Dynamic Array (arr4) - Allocate & Copy ---  
# arr4 allocated with size: 5  
# Copied arr1 into arr4.  
# Contents of arr4 (using foreach loop):  
#   arr4[0] = 0  
#   arr4[1] = 2  
#   arr4[2] = 4  
#   arr4[3] = 6  
#   arr4[4] = 8  
#  
# --- Task 6: Dynamic Array (arr4) - Resize ---  
# arr4 resized to: 7  
# Contents of arr4 after resize (using for loop):  
#   arr4[0] = 0  
#   arr4[1] = 2  
#   arr4[2] = 4  
#   arr4[3] = 6  
#   arr4[4] = 8  
#   arr4[5] = 99  
#   arr4[6] = 100  
#  
# --- Task 7: Dynamic Array (arr4) - Delete Last Element ---  
# arr4 size after deleting last element: 6  
#  
# --- Task 8: Dynamic Array (arr4) - Delete All ---  
# arr4 size after .delete(): 0  
# ** Note: $stop      : arrays_lab.sv(113)  
#   Time: 30 ns  Iteration: 0  Instance: /arrays_lab
```